TypeScript React Client Project — Grading Rubric

Instructor: Anthony Ackah-Nyanzu

1. Project Structure & Organization – 4 pts

Clean modular folder layout (`components`, `pages`, `features`, `store`, `services`, `types`, `hooks`). Use TypeScript types/interfaces. Proper separation of concerns.

2. Integration with API – 4 pts

Use Axios or Fetch to connect to the backend API. Handle JWT auth and render data dynamically with proper error handling.

3. Authentication & Authorization (Frontend) – 3 pts

Implement login/logout using JWT. Protect private routes using React Router. Show/hide UI elements based on user roles/permissions.

4. State Management (Redux Toolkit) – 3 pts

Use Redux Toolkit slices, actions, and selectors. Handle async operations with createAsyncThunk. Avoid prop drilling; maintain clean data flow.

5. UI/UX with Material UI – 3 pts

Consistent theme, responsive layout, professional design. Proper use of MUI components like AppBar, Dialog, Table, Snackbar.

6. Code Quality & TypeScript Usage – 2 pts

Clean, readable, fully typed TypeScript code. No major type or lint errors. Follow ESLint/Prettier rules if configured.

7. Documentation & Deployment – 1 pt

Include README with setup instructions and API connection details. Optional: deployed demo on Vercel/Netlify/GitHub Pages.

Category	Points
Project Structure & Organization	4
Integration with API	4
Authentication & Authorization	3
State Management (Redux Toolkit)	3
UI/UX with Material UI	3
Code Quality & TypeScript Usage	2
Documentation & Deployment	1
Total	20

Key Reminders for Students:
• Store JWT securely (in Redux state or memory, not cookies).
• Keep all API calls in a dedicated `services` folder.

- Use Redux Toolkit slices for state management and async logic.
- Maintain responsive design and consistent Material UI theming.
- Handle API errors gracefully and inform users (e.g., via Snackbars).
- Submit both source code and live demo link (if deployed).